

Ergonomics

<https://ehs.unc.edu/topics/ergonomics/>

Ergonomics can roughly be defined as the study of people in their working environment.

The word 'ergo' is derived from Latin, and means work. This is combined with the second half of the word (which is 'nomics') meaning laws.

When these two words are combined, it can be translated to the 'laws of work', insinuating that ergonomics shows the correct ways of working.

More specifically, an ergonomist (pronounced like economist) designs or modifies the work to fit the worker, not the other way around.

Ergonomics is the process of designing or arranging workplaces, products and systems so that they fit the people who use them.

Ergonomics is concerned with changing the environment (equipment, rate of work and bodily movements) to fit the requirements of individual employees rather than forcing them to adapt to environments that may not be suitable to their physical requirements and limitations.

The goal is to eliminate discomfort and risk of injury due to work.

Ergonomic Stressors

When evaluating a job, looking for three main characteristics known as ergonomic stressors:

- Uncomfortable postures - any awkward or static working postures adopted in completing a task
- Repetitive tasks - the repetitiveness of a task
- Force required to complete a task

Any of these factors, or any combination of these factors, may place someone at greater risk for discomfort.

Office Ergonomics

<https://ehs.unc.edu/topics/ergonomics/office-ergonomics/>

Many people don't realize that a poorly designed computer workstation and/or bad work habits can result in serious health problems.

Common symptoms associated with poor design or habits include discomfort in the back, neck and shoulders, hands and wrists, as well as headaches and eyestrain.

Musculoskeletal disorders (MSDs)

<https://ehs.unc.edu/topics/ergonomics/musculoskeletal-disorders/>

Musculoskeletal disorders (MSDs) are injuries or disorders of the muscles, nerves, tendons, joints, cartilage, and spinal discs.

Work-related musculoskeletal disorders (WMSDs) are conditions in which the work environment and performance of work contribute significantly to the condition.

Musculoskeletal disorders (MSDs), also called cumulative trauma disorders, are gradual-onset injuries that usually occur after repeated micro-trauma to a specific body part.

They may take weeks, months or years to develop and are often ignored at first due to the slow onset of symptoms.

MSDs most commonly affect the wrists, shoulders, neck, eyes and back. Specifically, they affect the nerves, tendons and muscles in those areas.

MSDs can also affect specific fingers, elbows and the knees, though these are less common.

Remember, MSDs are gradual-onset disorders, they usually take a while to occur.

Therefore, be on the lookout for lingering:

Discomfort or Pain: A tendon may not have nerve endings but if it starts to swell, becomes damaged or the muscles around it are being overworked, you may feel some discomfort or pain. If it occurs as soon as you arrive at work or start a task, if it wakes you up at night, follows you home, take notice!

Tingling, Numbness or Weakness: Some MSDs affect nerve function. If your hands “fall asleep” unexpectedly or wake you up in the middle of the night with that “pins and needles” feeling, this may be a sign of MSD onset. Likewise, if you experience trouble opening doors, jars, etc., this may be a sign.